

Design and Verification of an AMBA APB-Based GPIO Controller Using Verilog HDL

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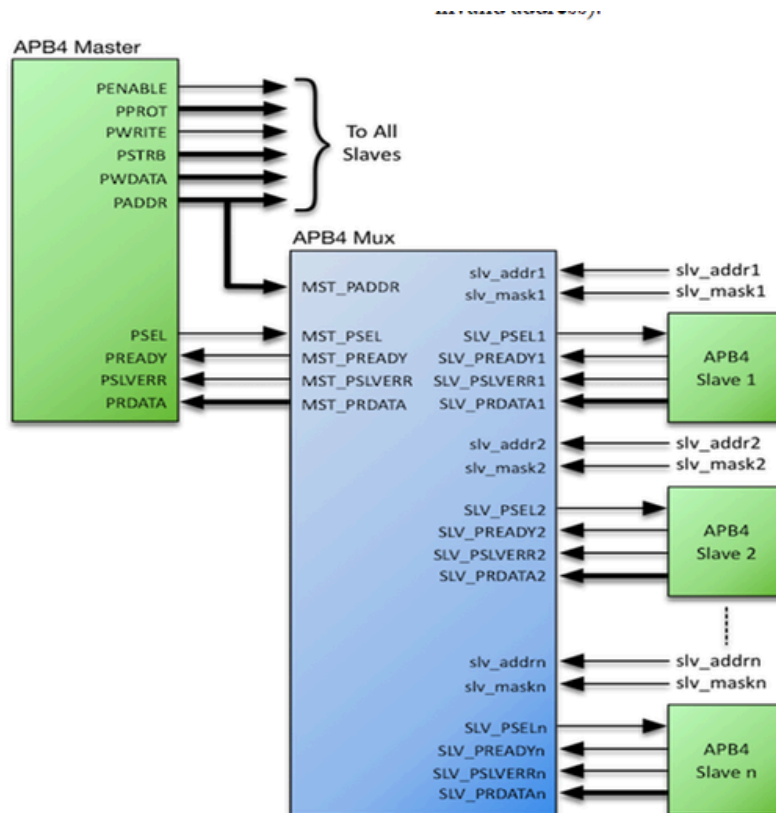
1. Introduction

GeneralPurposeInput/Output (GPIO) is a fundamental peripheral in modern System-on-Chip (SoC) designs, providing versatile digital signal interfaces for external communication. The Advanced Peripheral Bus (APB) is a low-power, reduced-complexity bus protocol within the AMBA hierarchy, specifically optimized for connecting peripheral devices to the system bus. Utilizing APB for GPIO control ensures efficient data transfer with minimal power overhead and straightforward interfacing logic. The objective of this project is to design, implement, and verify an 8-bit APB-compliant GPIO controller using Verilog HDL, focusing on memory-mapped register accessfor direction control and data manipulation.

2. Project Specifications

Item	Value
Protocol	AMBA APB
Language	Verilog HDL
Simulator	QuestaSim
GPIO Width	8-bit
Address Width	8-bit
Data Width	32-bit

3. AMBA APB Protocol Overview



The Advanced Peripheral Bus (APB) is designed for low-bandwidth control accesses. It uses a simple handshake involving PSEL, PENABLE, and PREADY to ensure reliable data transfer between the master and slow peripherals.

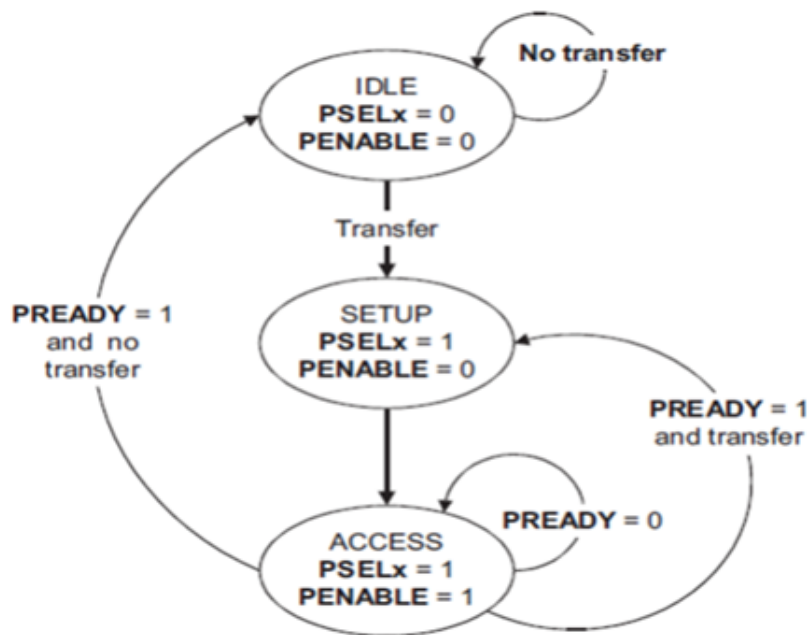
4. APB Signals

Signal	Width	Description
PCLK	1-bit	Clock signal for synchronization.
PRESETn	1-bit	Active-low reset signal.
PSEL	1-bit	Selects the peripheral.
PENABLE	1-bit	Indicates the access phase.
PWRITE	1-bit	HIGH for Write, LOW for Read.
PADDR	8-bit	Address bus.
PWDATA	32-bit	Write data bus.
PRDATA	32-bit	Read data bus.
PREADY	1-bit	Slave ready handshake.

5. Register Map

Address	Register	Function
0x00	DATA	GPIO Data (Input sample / Output drive)
0x04	DIRECTION	GPIO Direction (0: Input, 1: Output)

6. State Machine (FSM)



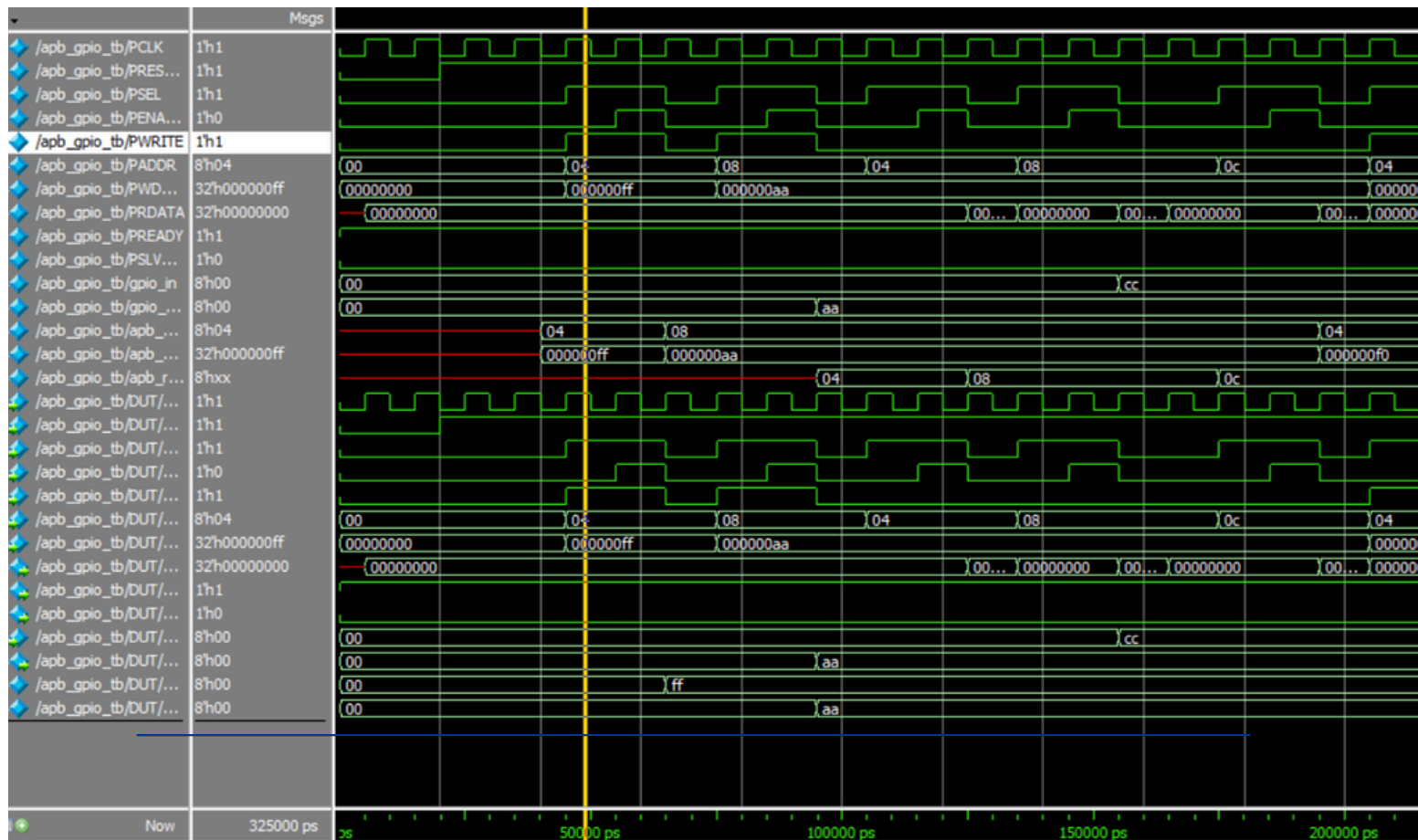
- **IDLE:** Peripheral is inactive; PSEL and PENABLE are low.
- **SETUP:** One-cycle state where PSEL is high to select the slave.
- **ACCESS:** PENABLE is high; data transfer completes when PREADY is high.

7. RTL Design Details

The RTL implementation ensures reliable register-to-pin mapping and protocol Compliance.

- **Reset Logic:** Asynchronous active-low reset initializes all registers to zero.
- **Write Logic:** Synchronous write to DATA or DIRECTION registers occurs during valid access
- **Read Logic:** Multiplexed PRDATA drives register contents or pin status back to the master.
- **GPIO Logic:** Tri-state buffer control allows pins to be driven as outputs or sampled as inputs.

8. Verification & Waveforms



```

# Errors: 0, Warnings: 0
# vsim -voptargs="" +acc" work.apb_gpio_tb
# Start time: 12:41:32 on Jul 07, 2026
# ** Note: (vsim-8009) Loading existing optimized design _opt
# Loading work.apb_gpio_tb(fast)
# Loading work.apb_gpio(fast)
# TIME=0 PSEL=0 PENABLE=0 PWRITE=0 PADDR=00 PWDATA=00000000 PRDATA=xxxxxxxx GPIO_OUT=00000000 GPIO_IN=00000000
# TIME=5000 PSEL=0 PENABLE=0 PWRITE=0 PADDR=00 PWDATA=00000000 PRDATA=00000000 GPIO_OUT=00000000 GPIO_IN=00000000
#
# Writing GPIO Direction Register
# TIME=45000 PSEL=1 PENABLE=0 PWRITE=1 PADDR=04 PWDATA=000000ff PRDATA=00000000 GPIO_OUT=00000000 GPIO_IN=00000000
# TIME=55000 PSEL=1 PENABLE=1 PWRITE=1 PADDR=04 PWDATA=000000ff PRDATA=00000000 GPIO_OUT=00000000 GPIO_IN=00000000
#
# Writing GPIO Output Register
# TIME=65000 PSEL=0 PENABLE=0 PWRITE=0 PADDR=04 PWDATA=000000ff PRDATA=00000000 GPIO_OUT=00000000 GPIO_IN=00000000
# TIME=75000 PSEL=1 PENABLE=0 PWRITE=1 PADDR=08 PWDATA=000000aa PRDATA=00000000 GPIO_OUT=00000000 GPIO_IN=00000000
# TIME=85000 PSEL=1 PENABLE=1 PWRITE=1 PADDR=08 PWDATA=000000aa PRDATA=00000000 GPIO_OUT=00000000 GPIO_IN=00000000
#
# Reading GPIO Direction Register
# TIME=95000 PSEL=0 PENABLE=0 PWRITE=0 PADDR=08 PWDATA=000000aa PRDATA=00000000 GPIO_OUT=10101010 GPIO_IN=00000000
# TIME=105000 PSEL=1 PENABLE=0 PWRITE=0 PADDR=04 PWDATA=000000aa PRDATA=00000000 GPIO_OUT=10101010 GPIO_IN=00000000
# TIME=115000 PSEL=1 PENABLE=1 PWRITE=0 PADDR=04 PWDATA=000000aa PRDATA=00000000 GPIO_OUT=10101010 GPIO_IN=00000000
# -----
# Time = 125000
# Read Address = 04
# Read Data = 00000000
# -----
#
# Reading GPIO Output Register
# TIME=125000 PSEL=0 PENABLE=0 PWRITE=0 PADDR=04 PWDATA=000000aa PRDATA=000000ff GPIO_OUT=10101010 GPIO_IN=00000000
# TIME=135000 PSEL=1 PENABLE=0 PWRITE=0 PADDR=08 PWDATA=000000aa PRDATA=00000000 GPIO_OUT=10101010 GPIO_IN=00000000
# TIME=145000 PSEL=1 PENABLE=1 PWRITE=0 PADDR=08 PWDATA=000000aa PRDATA=00000000 GPIO_OUT=10101010 GPIO_IN=00000000
# -----
# Time = 155000
# Read Address = 08
# Read Data = 00000000
# -----
#
# TIME=155000 PSEL=0 PENABLE=0 PWRITE=0 PADDR=08 PWDATA=000000aa PRDATA=000000aa GPIO_OUT=10101010 GPIO_IN=11001100
# TIME=165000 PSEL=0 PENABLE=0 PWRITE=0 PADDR=08 PWDATA=000000aa PRDATA=00000000 GPIO_OUT=10101010 GPIO_IN=11001100

```

```

#
# Reading GPIO Output Register
# TIME=125000 PSEL=0 PENABLE=0 PWRITE=0 PADDR=04 FWDATA=000000aa PRDATA=000000ff GPIO_OUT=10101010 GPIO_IN=00000000
# TIME=135000 PSEL=1 PENABLE=0 PWRITE=0 PADDR=08 FWDATA=000000aa PRDATA=00000000 GPIO_OUT=10101010 GPIO_IN=00000000
# TIME=145000 PSEL=1 PENABLE=1 PWRITE=0 PADDR=08 FWDATA=000000aa PRDATA=00000000 GPIO_OUT=10101010 GPIO_IN=00000000
# -----
# Time = 155000
# Read Address = 08
# Read Data = 00000000
# -----
# TIME=155000 PSEL=0 PENABLE=0 PWRITE=0 PADDR=08 FWDATA=000000aa PRDATA=000000aa GPIO_OUT=10101010 GPIO_IN=11001100
# TIME=165000 PSEL=0 PENABLE=0 PWRITE=0 PADDR=08 FWDATA=000000aa PRDATA=00000000 GPIO_OUT=10101010 GPIO_IN=11001100
#
# Reading GPIO Input Register
# TIME=175000 PSEL=1 PENABLE=0 PWRITE=0 PADDR=0c FWDATA=000000aa PRDATA=00000000 GPIO_OUT=10101010 GPIO_IN=11001100
# TIME=185000 PSEL=1 PENABLE=1 PWRITE=0 PADDR=0c FWDATA=000000aa PRDATA=00000000 GPIO_OUT=10101010 GPIO_IN=11001100
# -----
# Time = 195000
# Read Address = 0c
# Read Data = 00000000
# -----
#
# Changing GPIO Direction Register
# TIME=195000 PSEL=0 PENABLE=0 PWRITE=0 PADDR=0c FWDATA=000000aa PRDATA=000000cc GPIO_OUT=10101010 GPIO_IN=11001100
# TIME=205000 PSEL=1 PENABLE=0 PWRITE=1 PADDR=04 FWDATA=000000f0 PRDATA=00000000 GPIO_OUT=10101010 GPIO_IN=11001100
# TIME=215000 PSEL=1 PENABLE=1 PWRITE=1 PADDR=04 FWDATA=000000f0 PRDATA=00000000 GPIO_OUT=10101010 GPIO_IN=11001100
# TIME=225000 PSEL=0 PENABLE=0 PWRITE=0 PADDR=04 FWDATA=000000f0 PRDATA=00000000 GPIO_OUT=10100000 GPIO_IN=11001100
# TIME=235000 PSEL=1 PENABLE=0 PWRITE=1 PADDR=08 FWDATA=000000aa PRDATA=00000000 GPIO_OUT=10100000 GPIO_IN=11001100
# TIME=245000 PSEL=1 PENABLE=1 PWRITE=1 PADDR=08 FWDATA=000000aa PRDATA=00000000 GPIO_OUT=10100000 GPIO_IN=11001100
# TIME=255000 PSEL=0 PENABLE=0 PWRITE=0 PADDR=08 FWDATA=000000aa PRDATA=00000000 GPIO_OUT=10100000 GPIO_IN=11001100
# -----
# GPIO Direction = F0
# GPIO Output Reg= AA
# Actual GPIO_OUT= 10100000

```

The waveform verifies correct APB setup and access phases, GPIO data register writes, direction register updates, and successful read transactions. The observed timing confirms compliance with the APB protocol.

9. Results & Conclusion

The project successfully demonstrates an APB-compliant GPIO peripheral. All simulation test cases (Reset, Write, Read, Direction control) resulted in a PASS, confirming functional correctness for digital design applications.